MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT DAY TANK AREA

DESIGN CALCULATION OF TANK FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2004

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
API 650, 12th Edition	Welded Tanks For Oil Storage

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project
Shall	Is to be understood as mandatory in relation to the requirements of this
Should	Is to be understood as a strong recommendation to comply with the
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1 INTENT

This document is prepared for design of Foundations of Day & LO Tank Area. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

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2 SCOPE

This document is prepared for design of Foundation for Day & LO Tanks in Power Plant Area. Design of concrete structures and foundations has been done in accordance with SBC-304, ACI-318M and API 650.

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3.0 Design of Foundation for HFO BUFFER TANK

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DOCUMENT NO. DBAF501379b)

Tank Loading Data				
	Loading Type			
HFO BUFFER TANK	Empyt (W)	85	kN	
	Operating (F1)	662	kN	68*0.992*9.81
	Max (Wt)	779	kN	80*0.992*9.81



Summary of Tank Specification

Net Volume of Tank		=	68 m3	(REFER DOCUMENT NO. DBAF501379b)
Gross Volume of Tank		=	80 m3	
Diameter of Tank, D		=	4.5 m	
Height of Tank, H		=	5.4 m	
Height of Storage Oil, H_L		=	4.2 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	3.85 m	
Outer Diameter of Concrete Ring Wall, OD		=	5.15 m	
Height of Concrete Ring Wall, h		=	2.10 m	
Thickness of Concrete Ring Wall, t		=	0.65 m	
	P.C.D of Bolts	=	4.7 m	
Top of Foundation		=	0.9 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	9.92 kN/m^3	
Unit Weight of Soil, γ_s		=	20 kN/m^3	
Unit Weight of Concrete, γ_c		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	41 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load	=	33 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	41.00	117.00
Seismic Loads	Normal	33.00	72.00

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Foundation Weight

a) Ring Wall Weight = 463.13 kN

a) Inside Soil Weight = 488.95 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 9.19 m^2

Full cross-sectional area for operating condition, Ao= 20.83 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 9.22 m^3

Sectional modulus for operating condition,Zo = 13.41 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 6.01 kN/m

So, Required Min Wall THK.= 0.4836 m = 483.6 mm < 650 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	747.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	51.00	41.00	117.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	747.00	41.00	117.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	747.00	23.10	50.40
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	448.20	23.10	50.40
	Maximun weight	LC106 (2.4.1-1)	D + F2	864.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	1699.08	0.00	81.57	81.57	6.13	OK
LC102	1003.08	203.10	131.18	87.13	5.07	OK
LC103	1699.08	203.10	96.71	66.42	6.88	OK
LC104	1699.08	98.91	88.94	74.19	7.48	OK
LC105	1400.28	98.91	74.60	59.85	8.91	OK
LC106	1816.08	0.00	87.18	87.18	5.74	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	1699.08	0.4	679.63	0.00	-	-
LC102	1003.08	0.4	401.23	41.00	9.8	OK
LC103	1699.08	0.4	679.63	41.00	16.6	OK
LC104	1699.08	0.4	679.63	23.10	29.4	OK
LC105	1400.28	0.4	560.11	23.10	24.2	OK
LC106	1816.08	0.4	726.43	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	1699.08	2.575	4375.13	0.00	-	-
LC102	1003.08	2.575	2582.93	203.10	12.7	OK
LC103	1699.08	2.575	4375.13	203.10	21.5	OK
LC104	1699.08	2.575	4375.13	98.91	44.2	OK
LC105	1400.28	2.575	3605.72	98.91	36.5	OK
LC106	1816.08	2.575	4676.40	0.00	-	-

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Ultimate Strength Design

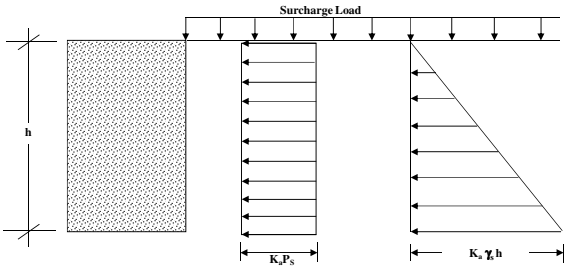
Load Combination (As per SBC-304 , Table 5.3.1)

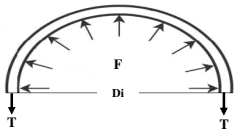
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	1045.80	0.00	15.90	8.95
Short term	Empty	LC102	0.9 D + 1.6*W	76.50	187.20	15.90	8.95
	Operating	LC103	1.2 (D + F1) + 1.6*W	896.40	187.20	15.90	8.95
		LC104	1.2 (D + F1) + E	896.40	72.00	15.90	8.95
		LC105	0.9 (D + F1) + E	672.30	72.00	15.90	8.95
	Test	LC106	1.4 (D + F2)	1209.60	0.00	15.90	8.95

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	1045.80	0.00	65.76
LC102	76.50	187.20	25.74
LC103	896.40	187.20	77.29
LC104	896.40	72.00	64.41
LC105	672.30	72.00	50.32
LC106	1209.60	0.00	76.05

Maximum Surcharge Load, Ps = 77.29 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 38.64 KN/m^2

P2 = $1.4 * \gamma_s * h * K_a$ = 29.40 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 112.02 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 215.64 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 570.5 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 3412.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2047.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

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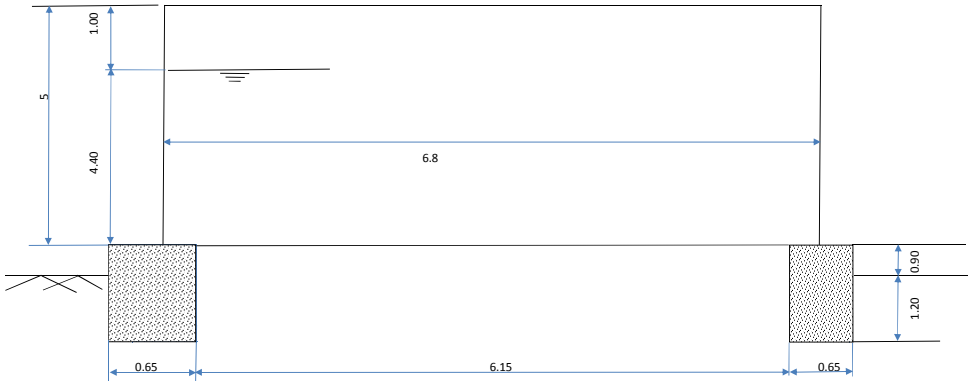
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4.0 Design of Foundation for LFO DAY TANK

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DOCUMENT NO. DBAF499260a)

Tank Loading Data				
	Loading Type			
LFO DAY TANK	Empty (W)	128	kN	
	Operating (F1)	1378	kN	156*0.9*9.81
	Max (Wt)	1607	kN	182*0.9*9.81



Summary of Tank Specification

Net Volume of Tank		=	156 m3	(REFER DOCUMENT NO. DBAF499260a)
Gross Volume of Tank		=	182 m3	
Diameter of Tank, D		=	6.8 m	
Height of Tank, H		=	5.4 m	
Height of Storage Oil, H_L		=	4.4 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	6.15 m	
Outer Diameter of Concrete Ring Wall, OD		=	7.45 m	
Height of Concrete Ring Wall, h		=	2.10 m	
Thickness of Concrete Ring Wall, t		=	0.65 m	
	P.C.D of Bolts	=	7 m	
Top of Foundation		=	0.9 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	9 kN/m^3	
Unit Weight of Soil, γ_S		=	20 kN/m^3	
Unit Weight of Concrete, γ_C		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	61 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load	=	64 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	61.00	179.00
Seismic Loads	Normal	64.00	132.00

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Foundation Weight

a) Ring Wall Weight = 699.85 kN

a) Inside Soil Weight = 1247.64 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 13.89 m^2

Full cross-sectional area for operating condition, Ao = 43.59 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 21.74 m^3

Sectional modulus for operating condition, Zo = 40.59 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 5.99 kN/m

So, Required Min Wall THK.= 0.5256 m = 525.6 mm < 650 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	1506.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	76.80	61.00	179.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	1506.00	61.00	179.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	1506.00	44.80	92.40
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	903.60	44.80	92.40
	Maximun weight	LC106 (2.4.1-1)	D + F2	1735.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	3453.49	0.00	79.22	79.22	6.31	OK
LC102	2024.29	307.10	159.90	131.66	4.16	OK
LC103	3453.49	307.10	86.79	71.66	7.66	OK
LC104	3453.49	186.48	83.82	74.63	7.93	OK
LC105	2851.09	186.48	70.00	60.81	9.50	OK
LC106	3682.49	0.00	84.48	84.48	5.92	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	3453.49	0.4	1381.39	0.00	-	-
LC102	2024.29	0.4	809.71	61.00	13.3	OK
LC103	3453.49	0.4	1381.39	61.00	22.6	OK
LC104	3453.49	0.4	1381.39	44.80	30.8	OK
LC105	2851.09	0.4	1140.43	44.80	25.5	OK
LC106	3682.49	0.4	1472.99	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	3453.49	3.725	12864.24	0.00	-	-
LC102	2024.29	3.725	7540.47	307.10	24.6	OK
LC103	3453.49	3.725	12864.24	307.10	41.9	OK
LC104	3453.49	3.725	12864.24	186.48	69.0	OK
LC105	2851.09	3.725	10620.30	186.48	57.0	OK
LC106	3682.49	3.725	13717.26	0.00	-	-

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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Ultimate Strength Design

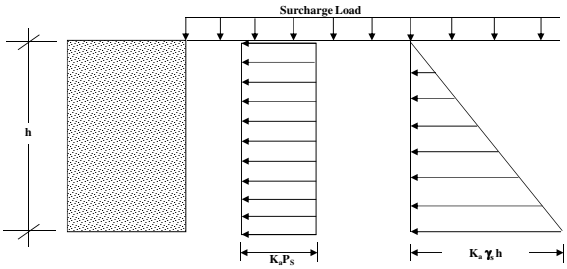
Load Combination (As per SBC-304 , Table 5.3.1)

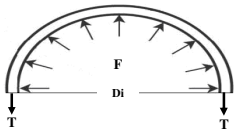
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	2108.40	0.00	36.32	30.87
Short term	Empty	LC102	0.9 D + W	115.20	179.00	36.32	30.87
	Operating	LC103	1.2 (D + F1) + W	1807.20	179.00	36.32	30.87
		LC104	1.2 (D + F1) + E	1807.20	132.00	36.32	30.87
		LC105	0.9 (D + F1) + E	1355.40	132.00	36.32	30.87
	Test	LC106	1.4 (D + F2)	2429.00	0.00	36.32	30.87

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	2108.40	0.00	58.06
LC102	115.20	179.00	8.97
LC103	1807.20	179.00	55.56
LC104	1807.20	132.00	54.04
LC105	1355.40	132.00	41.60
LC106	2429.00	0.00	66.88

Maximum Surcharge Load, Ps = 66.88 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 33.44 KN/m^2

P2 = $1.4 * \gamma_S * h * K_a$ = 29.40 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 101.10 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 310.88 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 822.4 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 3412.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2047.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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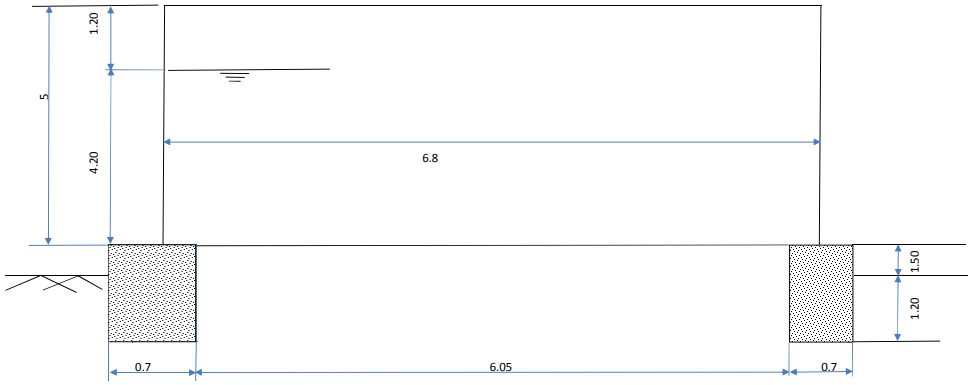
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5.0 Design of Foundation for HFO DAY TANK

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DOCUMENT NO. DBAF501381b)

Tank Loading Data				
	Loading Type			
HFO DAY TANK	Empyt (W)	140	kN	
	Operating (F1)	1510	kN	155*0.992*9.81
	Max (Wt)	1772	kN	182*0.992*9.81



Summary of Tank Specification

Net Volume of Tank		=	155 m3	(REFER DOCUMENT NO. DBAF501381b)
Gross Volume of Tank		=	182 m3	
Diameter of Tank, D		=	6.8 m	
Height of Tank, H		=	5.4 m	
Height of Storage Oil, H_L		=	4.2 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	6.05 m	
Outer Diameter of Concrete Ring Wall, OD		=	7.45 m	
Height of Concrete Ring Wall, h		=	2.7 m	
Thickness of Concrete Ring Wall, t		=	0.7 m	
	P.C.D of Bolts	=	7 m	
Top of Foundation		=	1.5 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	9.92 kN/m^3	
Unit Weight of Soil, γ_s		=	20 kN/m^3	
Unit Weight of Concrete, γ_c		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	63 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load	=	65 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	63.00	184.00
Seismic Loads	Normal	65.00	134.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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Foundation Weight

a) Ring Wall Weight = 961.89 kN

a) Inside Soil Weight = 1552.37 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 14.84 m^2

Full cross-sectional area for operating condition, Ao= 43.59 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 22.94 m^3

Sectional modulus for operating condition,Zo = 40.59 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 6.55 kN/m

So, Required Min Wall THK.= 0.6533 m = 653.3 mm < 700 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	1650.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	84.00	63.00	184.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	1650.00	63.00	184.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	1650.00	45.50	93.80
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	990.00	45.50	93.80
	Maximun weight	LC106 (2.4.1-1)	D + F2	1912.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	4164.26	0.00	95.53	95.53	5.23	OK
LC102	2598.26	354.10	190.47	159.60	3.49	OK
LC103	4164.26	354.10	104.25	86.81	6.38	OK
LC104	4164.26	216.65	100.87	90.19	6.59	OK
LC105	3504.26	216.65	85.73	75.05	7.76	OK
LC106	4426.26	0.00	101.54	101.54	4.92	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	4164.26	0.4	1665.70	0.00	-	-
LC102	2598.26	0.4	1039.30	63.00	16.5	OK
LC103	4164.26	0.4	1665.70	63.00	26.4	OK
LC104	4164.26	0.4	1665.70	45.50	36.6	OK
LC105	3504.26	0.4	1401.70	45.50	30.8	OK
LC106	4426.26	0.4	1770.50	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	4164.26	3.725	15511.87	0.00	-	-
LC102	2598.26	3.725	9678.52	354.10	27.3	OK
LC103	4164.26	3.725	15511.87	354.10	43.8	OK
LC104	4164.26	3.725	15511.87	216.65	71.6	OK
LC105	3504.26	3.725	13053.37	216.65	60.3	OK
LC106	4426.26	3.725	16487.82	0.00	-	-

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Ultimate Strength Design

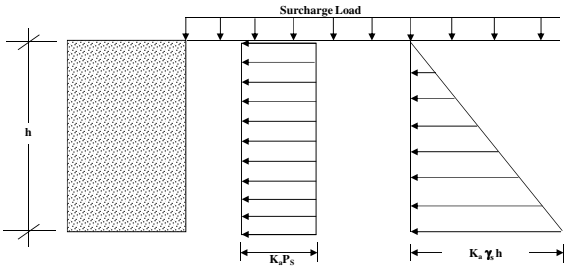
Load Combination (As per SBC-304 , Table 5.3.1)

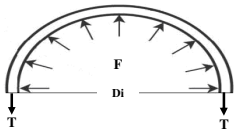
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	2310.00	0.00	36.32	30.87
Short term	Empty	LC102	0.9 D + W	126.00	184.00	36.32	30.87
	Operating	LC103	1.2 (D + F1) + W	1980.00	184.00	36.32	30.87
		LC104	1.2 (D + F1) + E	1980.00	134.00	36.32	30.87
		LC105	0.9 (D + F1) + E	1485.00	134.00	36.32	30.87
	Test	LC106	1.4 (D + F2)	2676.80	0.00	36.32	30.87

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	2310.00	0.00	63.61
LC102	126.00	184.00	9.43
LC103	1980.00	184.00	60.48
LC104	1980.00	134.00	58.86
LC105	1485.00	134.00	45.23
LC106	2676.80	0.00	73.71

Maximum Surcharge Load, Ps = 73.71 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 36.85 KN/m^2

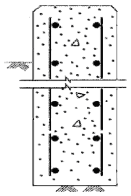
P2 = $1.4 * \gamma_s * h * K_a$ = 37.80 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 150.53 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 455.37 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 1204.7 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 4725 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2835 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
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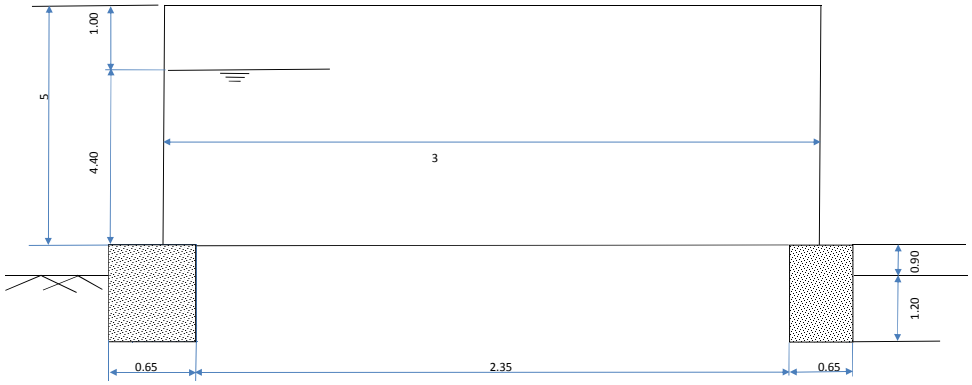
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6.0 Design of Foundation for SLUDGE TANK & LO STORAGE TANK

Since both the tanks have same capacity & same Operative weight, they are designed similarly.
(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DOCUMENT NO. DBAF501382a)

Tank Loading Data				
	Loading Type			
SLUDGE TANK	Empty (W)	47	kN	
	Operating (F1)	304	kN	31*1*9.81
	Max (Wt)	354	kN	36*1*9.81



Summary of Tank Specification

Net Volume of Tank		=	31 m3	(REFER DOCUMENT NO. DBAF501382a)
Gross Volume of Tank		=	36 m3	
Diameter of Tank, D		=	3 m	
Height of Tank, H		=	5.4 m	
Height of Storage Oil, H_L		=	4.4 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	2.35 m	
Outer Diameter of Concrete Ring Wall, OD		=	3.65 m	
Height of Concrete Ring Wall, h		=	2.10 m	
Thickness of Concrete Ring Wall, t		=	0.65 m	
	P.C.D of Bolts	=	3.2 m	
Top of Foundation		=	0.9 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	10 kN/m^3	
Unit Weight of Soil, γ_s		=	20 kN/m^3	
Unit Weight of Concrete, γ_c		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	27.5 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load	=	16.5 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	27.50	77.00
Seismic Loads	Normal	16.50	39.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Foundation Weight

a) Ring Wall Weight = 308.76 kN

a) Inside Soil Weight = 182.17 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 6.13 m^2

Full cross-sectional area for operating condition, Ao= 10.46 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 3.95 m^3

Sectional modulus for operating condition,Zo = 4.77 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 4.99 kN/m

So, Required Min Wall THK.= 0.3667 m = 366.7 mm < 650 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	351.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	28.20	27.50	77.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	351.00	27.50	77.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	351.00	11.55	27.30
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	210.60	11.55	27.30
	Maximun weight	LC106 (2.4.1-1)	D + F2	401.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	841.92	0.00	80.46	80.46	6.21	OK
LC102	519.12	134.75	118.82	50.66	5.60	OK
LC103	841.92	134.75	108.69	52.24	6.12	OK
LC104	841.92	51.56	91.26	69.66	7.29	OK
LC105	701.52	51.56	77.84	56.25	8.54	OK
LC106	891.92	0.00	85.24	85.24	5.87	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	841.92	0.4	336.77	0.00	-	-
LC102	519.12	0.4	207.65	27.50	7.6	OK
LC103	841.92	0.4	336.77	27.50	12.2	OK
LC104	841.92	0.4	336.77	11.55	29.2	OK
LC105	701.52	0.4	280.61	11.55	24.3	OK
LC106	891.92	0.4	356.77	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	841.92	1.825	1536.51	0.00	-	-
LC102	519.12	1.825	947.40	134.75	7.0	OK
LC103	841.92	1.825	1536.51	134.75	11.4	OK
LC104	841.92	1.825	1536.51	51.56	29.8	OK
LC105	701.52	1.825	1280.28	51.56	24.8	OK
LC106	891.92	1.825	1627.76	0.00	-	-

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Project Country	KINGDOM OF SAUDI ARABIA

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Ultimate Strength Design

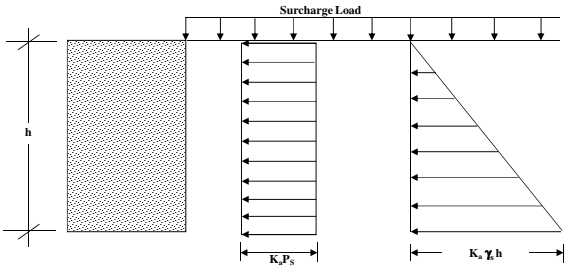
Load Combination (As per SBC-304 , Table 5.3.1)

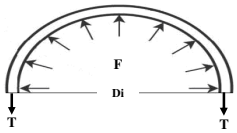
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	491.40	0.00	7.07	2.65
Short term	Empty	LC102	0.9 D + W	42.30	77.00	7.07	2.65
	Operating	LC103	1.2 (D + F1) + W	421.20	77.00	7.07	2.65
		LC104	1.2 (D + F1) + E	421.20	39.00	7.07	2.65
		LC105	0.9 (D + F1) + E	315.90	39.00	7.07	2.65
	Test	LC106	1.4 (D + F2)	561.40	0.00	7.07	2.65

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	491.40	0.00	69.52
LC102	42.30	77.00	35.03
LC103	421.20	77.00	88.64
LC104	421.20	39.00	74.30
LC105	315.90	39.00	59.40
LC106	561.40	0.00	79.42

Maximum Surcharge Load, Ps = 88.64 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 44.32 KN/m^2

P2 = $1.2 * \gamma_S * h * K_a$ = 25.20 KN/m^2 for LC103 - 1.2 (D + F1) + W

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 119.53 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 140.45 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = $T/(\phi \times Fy)$ = 371.5 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 3412.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2047.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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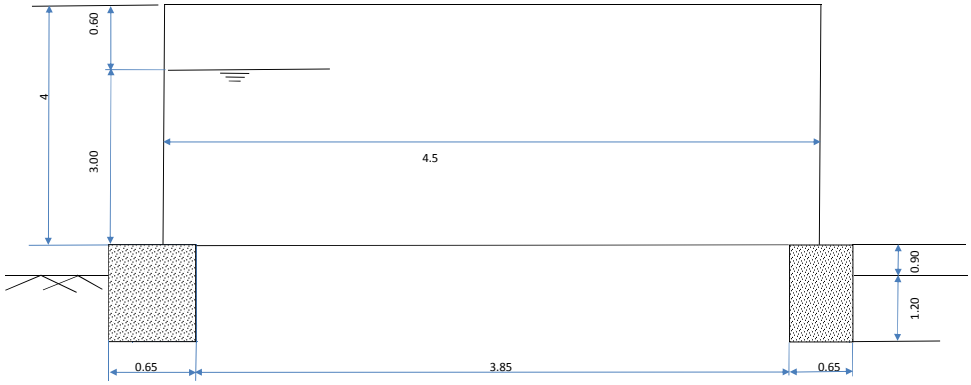
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7.0 Design of Foundation for OILY WATER BUFFER TANK

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (REFER DOC. NO. DBAF490344)

Tank Loading Data				
	Loading Type			
OILY WATER BUFFER TANK	Empty (W)	60	kN	
	Operating (F1)	500	kN	50*1*9.81
	Max (Wt)	500	kN	50*1*9.81



Summary of Tank Specification

Gross Volume of Tank		=	50 m3	(REFER DOC. NO. DBAF490344)
Diameter of Tank, D		=	4.5 m	
Height of Tank, H		=	3.6 m	
Height of Storage Oil, H_L		=	3 m	(Appox)
Inner Diameter of Concrete Ring Wall, ID		=	3.85 m	
Outer Diameter of Concrete Ring Wall, OD		=	5.15 m	
Height of Concrete Ring Wall, h		=	2.10 m	
Thickness of Concrete Ring Wall, t		=	0.65 m	
	P.C.D of Bolts	=	4.7 m	
Top of Foundation		=	0.9 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L		=	10 kN/m^3	
Unit Weight of Soil, γ_s		=	20 kN/m^3	
Unit Weight of Concrete, γ_c		=	24 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	27.5 kN	(REFER DOC. NO. DBAF490344)
Total Base Shear due to Seismic load	=	20 kN	

Load	**Load Condition**	**Base Shear (F,kN)**	**Moment (M,kN-m)**
Wind Loads	Normal	27.50	54.00
Seismic Loads	Normal	20.00	28.00

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Foundation Weight

a) Ring Wall Weight = 463.13 kN

a) Inside Soil Weight = 488.95 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 9.19 m^2

Full cross-sectional area for operating condition, Ao= 20.83 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 9.22 m^3

Sectional modulus for operating condition,Zo = 13.41 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 4.24 kN/m

So, Required Min Wall THK.= 0.6431 m = 643.1 mm < 650 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	560.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	36.00	27.50	54.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	560.00	27.50	54.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	560.00	14.00	19.60
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	336.00	14.00	19.60
	Maximun weight	LC106 (2.4.1-1)	D + F2	560.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{max}= V/A - M/Z kN/m^2	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	1512.08	0.00	72.59	72.59	6.89	OK
LC102	988.08	111.75	119.65	95.41	5.56	OK
LC103	1512.08	111.75	80.92	64.26	8.22	OK
LC104	1512.08	49.00	76.24	68.93	8.72	OK
LC105	1288.08	49.00	65.49	58.18	10.15	OK
LC106	1512.08	0.00	72.59	72.59	6.89	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	1512.08	0.4	604.83	0.00	-	-
LC102	988.08	0.4	395.23	27.50	14.4	OK
LC103	1512.08	0.4	604.83	27.50	22.0	OK
LC104	1512.08	0.4	604.83	14.00	43.2	OK
LC105	1288.08	0.4	515.23	14.00	36.8	OK
LC106	1512.08	0.4	604.83	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	1512.08	2.575	3893.60	0.00	-	-
LC102	988.08	2.575	2544.30	111.75	22.8	OK
LC103	1512.08	2.575	3893.60	111.75	34.8	OK
LC104	1512.08	2.575	3893.60	49.00	79.5	OK
LC105	1288.08	2.575	3316.80	49.00	67.7	OK
LC106	1512.08	2.575	3893.60	0.00	-	-

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Ultimate Strength Design

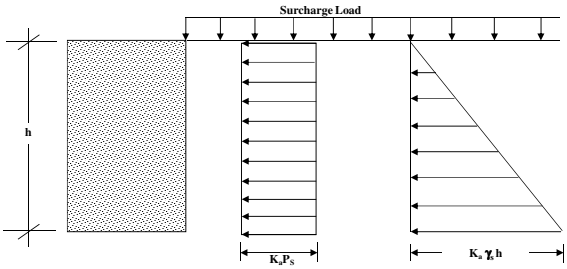
Load Combination (As per SBC-304 , Table 5.3.1)

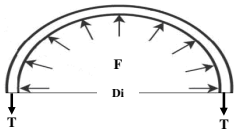
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	784.00	0.00	15.90	8.95
Short term	Empty	LC102	0.9 D + W	54.00	54.00	15.90	8.95
	Operating	LC103	1.2 (D + F1) + W	672.00	54.00	15.90	8.95
		LC104	1.2 (D + F1) + E	672.00	28.00	15.90	8.95
		LC105	0.9 (D + F1) + E	504.00	28.00	15.90	8.95
	Test	LC106	1.4 (D + F2)	784.00	0.00	15.90	8.95

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	784.00	0.00	49.29
LC102	54.00	54.00	9.43
LC103	672.00	54.00	48.29
LC104	672.00	28.00	45.38
LC105	504.00	28.00	34.82
LC106	784.00	0.00	49.29

Maximum Surcharge Load, Ps = 49.29 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 24.65 KN/m^2

P2 = $1.4 * \gamma_S * h * K_a$ = 29.40 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 82.63 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 159.06 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 420.8 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 3412.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2047.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 5 bar D20 (@ 200mm)

=3142mm2 > Required As OK

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8.0 Design of Foundation for LO STORAGE TANK (USED)

Summary of Tank Specification

Gross Volume of Tank	=	20	m3	(REFER DOCUMENT NO. DBAF490344)
Operative Weight of Tank	=	226	kN	(REFER DWG NO. MD-216-49PR-EG-EL-LAD-V553-1002)
Diameter of Tank, D	=	2	m	
Length of Tank, H	=	6.96	m	
Height of tank centre from pedestal top	=	1.5	m	(assumed)
Length of Pedestal	=	2.25	m	
Width of Pedestal	=	0.4	m	
Length of Foundation	=	2.75	m	
Width of Foundation	=	1.25	m	
Top of Foundation above FGL	=	1	m	
Thickness of Foundation	=	0.25	m	
Bottom of foundation below FGL	=	1.2	m	
Unit Weight of Soil, γ_s	=	20	kN/m^3	
Unit Weight of Concrete, γ_c	=	24	kN/m^3	
Base area of Foundation =	=	3.44	m^2	Section Modulus of Base = 1.01 m^3
Weight of Soil above foundation = (2.75-2.25)*(1.25-0.4)*20*1.2	=	10.2	kN	
Weight of foundation = (2.25*0.4*1.95+2.75*1.25*0.25) *24	=	62.745	kN	
Total weight of soil + foundation =		72.945	kN	

Calculation of Seismic Load (SL) for LO Storage Tank :

Seismic Parameters :

Seismic Design Category =	B		
Site Class =	B		
Seismic Coeffecient, Ss =	0.1	g	(From Civil-Structural Design Criteria)
S_1 =	0.02	g	
Occupancy importance factor, I =	1		(From Table 9.5, SBC 301:2007)
Response modification factor, R =	4		(From Table 10.2, SBC 301:2007)
Height of the structure, h =	4	m	approx.
For steel frames,			
Time Period, T = $0.055 * h^{0.75}$ =	0.16	sec	(eqn. 10.9.3.2-1 & Table 10.9.3.2, SBC 301:2007)
Factors, Fa =	1		(From Table 9.4.3a, SBC 301:2007)
Fv =	1		(From Table 9.4.3b, SBC 301:2007)
Maximum spectral acceleration, S_{MS} = Fa * Ss =	0.1	g	(eqn. 9.4.3-1, SBC 301:2007)
Maximum spectral acceleration, S_{M1} = Fa * S_1 =	0.02	g	(eqn. 9.4.3-2, SBC 301:2007)
Design spectral acceleration, S_{DS} = 2/3 * S_{MS} =	0.067	g	(eqn. 9.4.4-1, SBC 301:2007)
Design spectral acceleration, S_{D1} = 2/3 * S_{M1} =	0.013	g	(eqn. 9.4.4-2, SBC 301:2007)
Seismic base Shear, V =	Cs * W		(eqn. 10.9.2-1, SBC 301:2007)
Seismic Response co-eff, Cs = S_{DS} * I /R =	0.017	g	(eqn. 10.9.2.1-1, SBC 301:2007)
Max. value of C_S = S_{D1} *I /(T * R) =	0.021	g	(eqn. 10.9.2.1-2, SBC 301:2007)
Min. value of C_S = 0.044 * S_{DS} * I =	0.003	g	(eqn. 10.9.2.1-3, SBC 301:2007)
Total Load , W =	226	kN	(Operative Weight of Tank)
Seismic base Shear, V =	**36.96**	kN	
Moment at foundation base =	**136.76**	**kN-m**	

Stability Check

Restoring force against sliding = (226+72.945)*0.4 =	119.578	kN	
Factor of Safety against sliding = 0.9*119.578/ (36.96)=	**2.9 > 1.5 - Hence OK**		
Restoring Moment against Overtuening = 0.9*(226+72.945)*2.75/2 =	369.95	kN-m	
Factor of Safety against overturning = 0.9*369.95/ (136.76)=	**2.4 > 1.5 - Hence OK**		

Allowable Bearing Capacity of Soil, Q_{all} =	=	500	KN/m^2	
Gross Pressure at base level of foundation = (226+72.945) /(2*3.4375) + (136.76/(2*1.01))	=	111.19	KN/m^2	
Net Pressure at foundation base	=	87.19	KN/m^2	OK - < 500kN/m2
Minimum Reinforcement in Pedestal = 0.005 * 2.25*0.4*10^6	=	4500	mm^2	
Vertical Reinforcement provided = **24 no.** **16** **dia bars =**		**4825**	**mm^2**	
Minimum Reinforcement in Foundation = 0.0025 * 0.25*1000	=	625	mm^2/m	
Reinforcement provided = **16 dia bars @** **200** **c/c =**	=	**1005**	**mm^2/m**	

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	30-07-2020	24-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	30-07-2020	24-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

REFERENCE

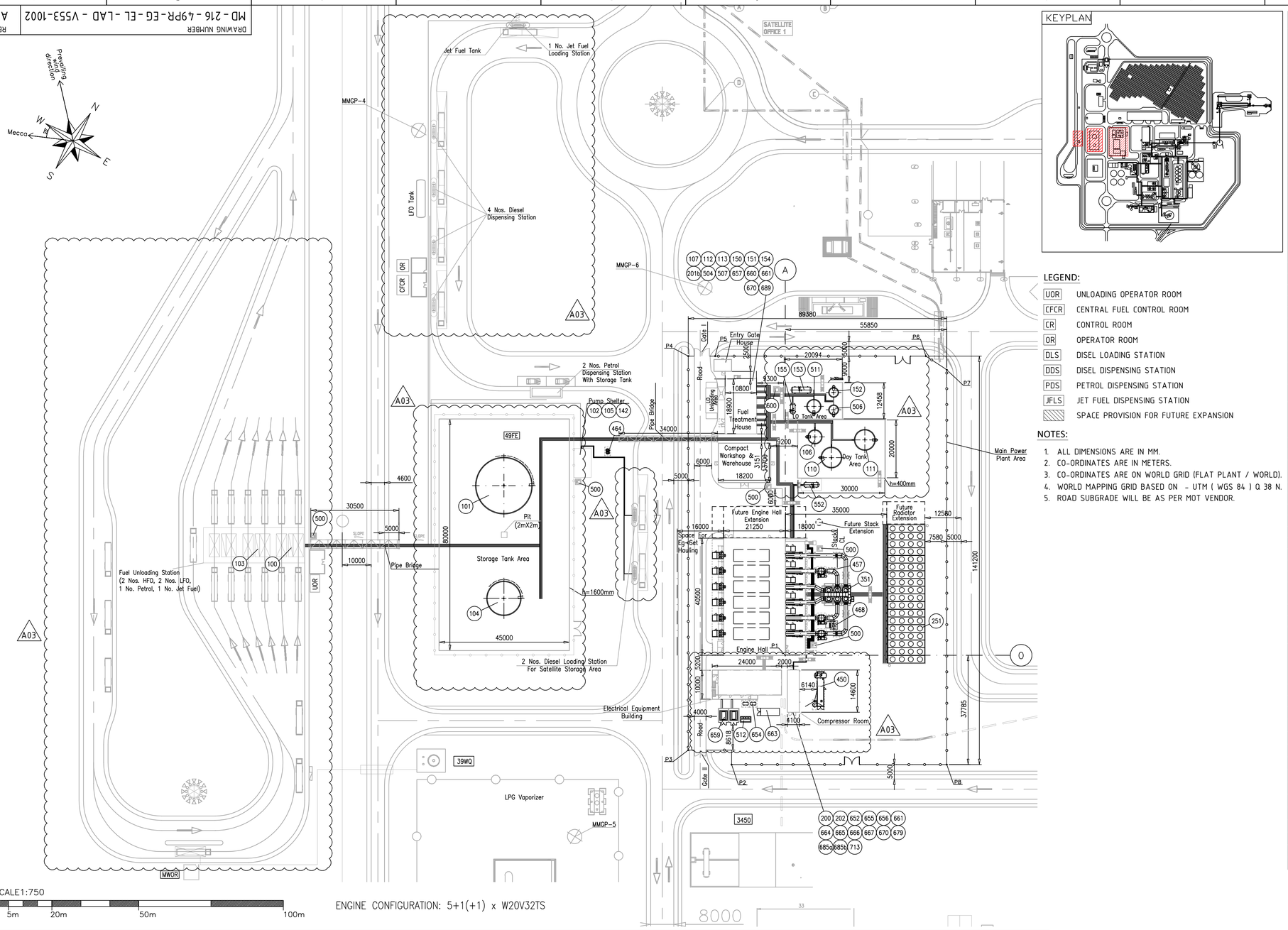
[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] API 650 , 12th Edition
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	30-07-2020	24-08-2020	26-10-2023				Date	30-07-2020	24-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 POWER GENERATION LAYOUT DRAWING

DRAWING NUMBER: MD - 216 - 49PR - EG - EL - LAD - V553-1002 | REV: A03



KEYPLAN

LEGEND:

UOR	UNLOADING OPERATOR ROOM
CFCR	CENTRAL FUEL CONTROL ROOM
CR	CONTROL ROOM
OR	OPERATOR ROOM
DLS	DISEL LOADING STATION
DDS	DISEL DISPENSING STATION
PDS	PETROL DISPENSING STATION
JFLS	JET FUEL DISPENSING STATION
	SPACE PROVISION FOR FUTURE EXPANSION

NOTES:

1. ALL DIMENSIONS ARE IN MM.
2. CO-ORDINATES ARE IN METERS.
3. CO-ORDINATES ARE ON WORLD GRID (FLAT PLANT / WORLD).
4. WORLD MAPPING GRID BASED ON - UTM (WGS 84) Q 38 N.
5. ROAD SUBGRADE WILL BE AS PER MOT VENDOR.

SITE AREA (This list applies to Engine Hall plan, Engine hall section and Site layouts)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
251	18	VCA	Radiator (NL2)		6130	+5.000
351	6	NHA	Exhaust gas silencer		4790	+4.960
450	1	RHC	Heat recovery container		~19000	–
457	3	RCC	Exhaust gas boiler		17800	+0.221
464	1	RDB	Condensate tank		1350	–
468	1	DAD	Boiler washing water tank	5	~5620	–
500	6	DAA	Oily water collecting pit	2.5	–	–
512	1	–	Control pit		–	–
654	2	BAZ	Earthing transformer		–	–
659	2	BFB	Station transformer		–	–
663	1	BLM	Black starting unit		~5500	–

DAY / STORAGE TANK / UNLOADING AREA (This list applies to Power plant site layout)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
100	2	PAA	HFO Unloading Pump (double)	–	1242	+0.200
101	1	PAB	HFO Storage Tank (frangible roof joint)	4000	4107000	+0.400
103	2	PAD	LFO Unloading Pump (double)		995	+0.200
104	1	PAE	LFO Storage Tank	1000	1042900	+0.400
106	1	PBA	HFO Buffer Tank	80	87800	+0.400
110	1	PBC	HFO Day Tank	150	163400	+0.500
111	1	PBF	LFO Day tank	150	163000	+0.400
152	1	QAC	LO Storage Tank (Clean)	35	39600	+0.400
153	1	QAD	LO Storage Tank (Used)	20	22590	+0.500
155	1	QAM	LO Service Tank	7	8090	+0.500
506	1	DDB	Sludge Tank	35	39700	+0.400
511	1	DAB	Oily Water Buffer Tank	50	56000	+0.400
552	1	VBC	Treated water tank	10	11450	–

FUEL TREATMENT HOUSE / PUMP SHELTER

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
102	1	PAC	HFO Transfer Pump Unit (double)		2645	±0.000
105	1	PAF	LFO Transfer Pump Unit (double)		486	±0.000
107	1	PBB	HFO Separator Unit (double)		4900	±0.000
112	1	PCA	HFO Feeder 3.0 (double)		3100	±0.000
113	1	PCA	LFO Feeder 3.0 (double)		1466	±0.000
142	2	–	LFO Loading Pump (double)		955	–
150	1	QAA	LO Unloading Pump Unit, (clean)		264	±0.000
151	1	QAB	LO Loading Pump Unit, (used)		264	±0.000
154	1	QAE	LO Transfer Pump Unit		665	±0.000
201b	1	TCC	Instrument air bottle	0.2	–	±0.000
504	1	DBB	Oily Water Treatment Unit		~11500	±0.000
507	1	–	Sludge Transfer Pump Unit		97	±0.000
657	1	BFA	LV Switchgear, Fuel treatment house		–	–
660	1	BJG	HFO Tank Contr Panel + Remote Commo		–	–
661	1	BLI	Lighting Panel *		–	–
670	3	–	Frequency Converters for Feeder Pumps		–	–
689	2	–	Frequency Converter for HFO Separator		–	–

ELECTRICAL EQUIPMENT BUILDING (This list applies to Engine Hall plan, Engine hall section and Site layouts.)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Incl. liquids	Mounting level (Floor level)
200	2	TCA	Instrument and working air unit		570	+2.100
202	2	TSA	Starting air unit		1200	+2.100
652	1	BAC	MV Switchgear			+2.100
655	2	BEY	DC-panel			+2.100
656	1	BFA	LV Switchgear			+2.100
661	1	BLI	Lighting panel			+2.100
664	1	BLN	Fire detection panel			+2.100
665	1	BLQ	Ventilation panel			+2.100
666	1	CFA	Control panel, common			+2.100
667	6	CFC	Control panel, engine wise			+2.100
670	6	BLP	Frequency converter, (Radiator)			+2.100
679	1		Desk for Phone/Printer/PC			+2.100
685a	1		Server rack		135	+2.100
685b	1		Communication cabinet			+2.100
713	3		Ventilation outlet fan			–

SITE INFO (This list applies to Engine Hall plan, section and Power lant site layouts)

FFL	±0.000 m	(+913.6 m from sea level)
FGL	−0.500 m	(+913.1 m from sea level)
RL	−0.300 m	(+913.3 m from sea level)

LEGEND (This list applies to Power plant site layout)

Colour:
- Items in customer scope
- Items in Wärtsilä scope

Linetype:
- Foundation/hidden structure
- Engine foundations
- Road center line
- Fence line
- Item above ground
- Site fence for main Power plant area
- Future extension

COORDINATES – UTM (WGS 84) Q38 N

No.	(E)x	(N)y	Description
P1	263987.528	2506005.175	A/0 point of Engine hall
P2	263991.831	2505963.310	Fence post
P3	263976.470	2505959.607	Fence post
P4	263904.291	2506075.108	Fence post
P5	263912.170	2506080.032	Fence post
P6	263973.959	2506118.642	Fence post
P7	263982.571	2506118.482	Fence post
P8	264054.918	2506002.731	Fence post

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A03	01/06/2020	REVISED AS PER COMMENTS DATED 13/05/2020	--	SJK	--	NBR	
A02	17/04/2020	REVISED AS PER COMMENTS DATED 16/03/2020	--	SJK	--	NBR	
A01	14/02/2020	ISSUED FOR APPROVAL	--	SJK	--	NBR	

QRS

DOCUMENT NUMBER | REFERENCE DOCUMENTS

WÄRTSILÄ

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Energy Business

DRAWING NUMBER	DBAF489643				REVISION	f
DATE	29-05-2020	DATE	29-05-2020	DATE	29-05-2020	
DRAWN	hslx01	CHECKED	mspx03	APPROVED	mnyx02	

LARSEN & TOUBRO DRAWING NUMBER: O19045_008_M_V_12021_001 | REVISION: 2

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __WG01__MTL01 __45003 | DOCUMENT ID: OU100673503 | REVISION: 2

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Saudi Arabian Mining Company شركة معادن العربية السعودية
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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SJK	12/02/2020	LAYOUT
ENGINEER	--	--	POWER GENERATION
APPROVED	NBR	14/02/2020	PLAN
EMPLOYER			
SCALE	1:750	A1	DRAWING NUMBER: MD - 216 - 49PR - EG - EL - LAD - V553-1002 | REV: A03

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	**D01**				Outotec / L&T Revision No.:	0	1	2			
Date	30-07-2020	24-08-2020	**26-10-2023**				Date	30-07-2020	24-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 GA DRAWING OF HFO BUFFER TANK

Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title:	**HFO Buffer Tank** **49PR POWER GENERATION** **Dimensional drawing**

Document Number:
Revision:
Revision Date:
Status:

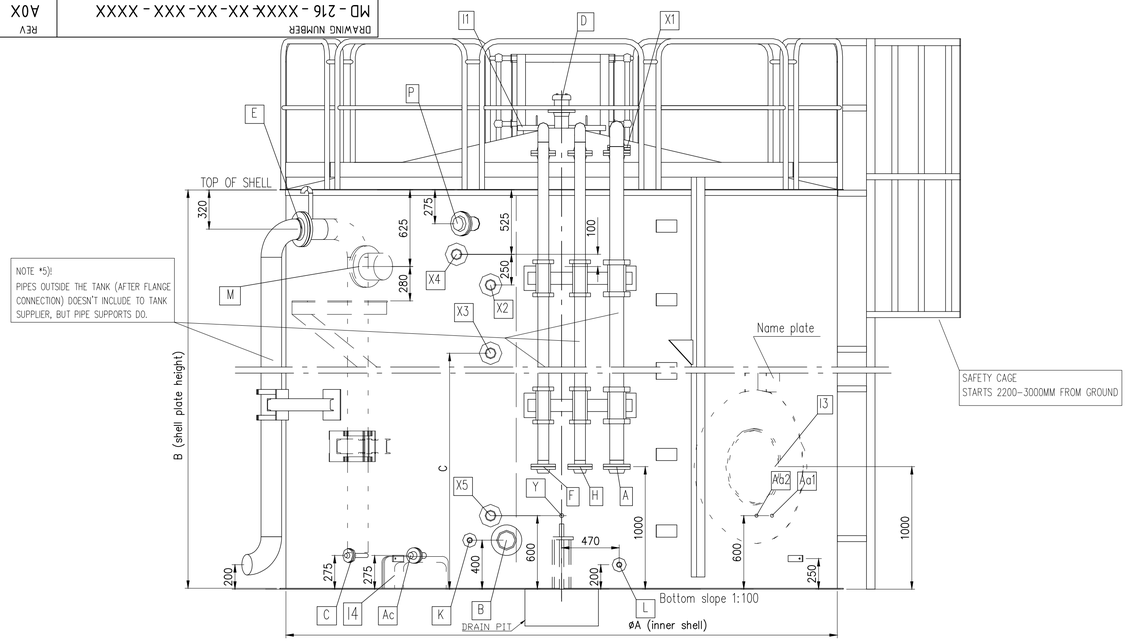
Outotec Document Number:							
L&T Document Number:							

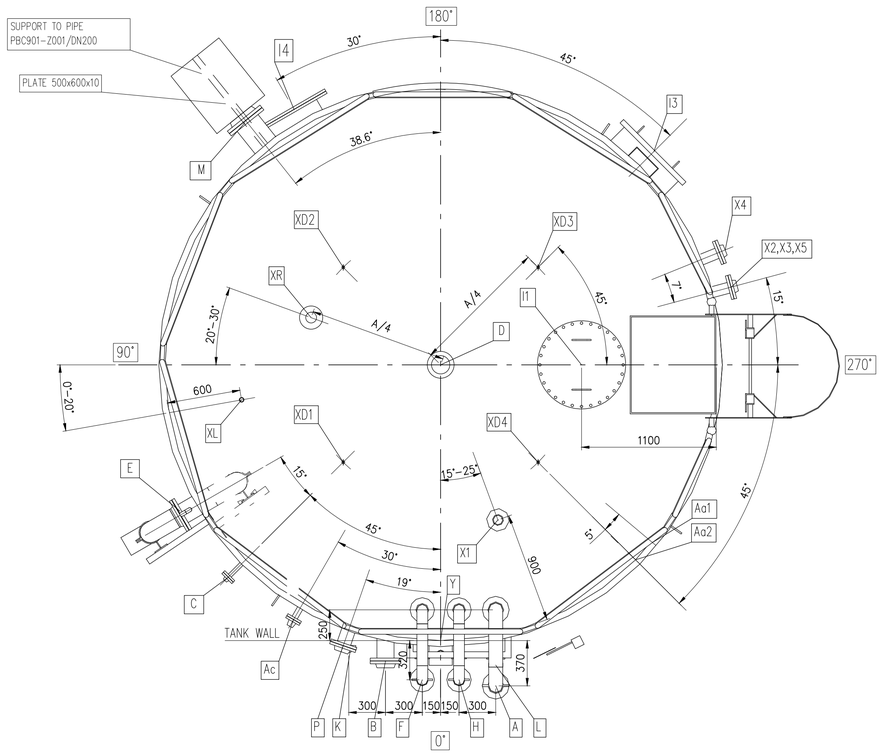
WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF501379 b
	Original Doc no:
	Wartsila Supplier:

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer

DRAWING NUMBER MD-216-XXXX-XX-XX-XXX-XXXX REV A0X





NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	HFO FILLING	4" / DN100	x	450 FROM 0°	150	ROOF	SEE NOTE *5)
B	SUCTION TO UNIT	5" / DN125	x	450 FROM 0°	200	400	
C	DRAIN / EMPTYING	1" / DN25	x	45°	150	275	
D	VENTING PIPE / FLAME ARRESTER	4", 150lbs	x	AT CENTER	150	ROOF	
E	OVERFLOW	6" / DN150	x	60°	200	320 FROM T.O.S	SEE NOTE *5) & DBAB685158
F	RETURN FROM UNIT	3" / DN80	x	150 FROM 0°	150	ROOF	SEE NOTE *5)
H	RETURN FROM UNIT	3" / DN80	x	150 FROM 0°	150	ROOF	SEE NOTE *5)
I1	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	x API 650	270°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	x API 650	225°	150	1000	
I4	CLEANOUT DOOR	203 X 406		150°		0	See Table 5.9a, 5.10a, 5.11a in API650
K	STEAM INLET	1" / DN25	x	750 FROM 0°	150	400	MINIMUM COIL LENGTH 16.8M
L	CONDENSATE OUTLET	1" / DN25	x	470 FROM 0°	150	200	
Y	STEAM CONTROL THERMOSTAT	G1"		0°		600	
M	OVERFLOW FROM DAY TANK	8" / DN200	x	141.4°	300	625 FROM T.O.S.	NOTE, SUPPORT TO PIPE PBC901-Z001/DN200
Aa1	TEMPERATURE LOCAL	G1/2"		310°		600	
Aa2	TEMPERATURE SENSOR	G1/2"		315°		600	
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	4" / DN100	x	19°	175	275 FROM T.O.S.	
X2	HIGH LEVEL CONTROL	3" (150 lbs)	x	255°	175 MAX	775 FROM T.O.S.	
X3	LOW LEVEL CONTROL	3" (150 lbs)	x	255°	175 MAX	C	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	x	248°	175 MAX	525 FROM T.O.S.	
X5	LOW LOW LEVEL CONTROL	3" (150 lbs)	x	255°	175 MAX	600	
Ac	CONN.FOR PRESSURE TRANSM. (OPT./BLIND FL.)	1 1/2"(150 lbs)	x	30°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR	–	x	335°-345°	150	ROOF	
XR	CONN. FOR LEVEL MEASUREMENT	3" (150 lbs)	x	110°-120°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL (PLUGGED)	G1"		70°-90°		ROOF	
XD1-XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14 NPT		45°, 135°, 225°, 315°		ROOF	TOTAL 4 PCS
x ADDITIONAL INFO FROM API 650 DATA SHEETS							

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
LADDERS	–	1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000		1 SET	
INSULATION	S=100mm	1 SET	TANK SHELL TO BE INSULATED
HEATING COIL	DN25	1 SET	MINIMUM COIL LENGTH 16.8M
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x ADDITIONAL INFO FROM API 650 DATA SHEETS			

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS

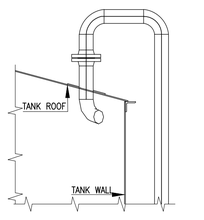
MEASURES				VOLUMES		PRELIMINARY DESIGN VALUES		
Nominal size	øA (mm)	B (mm)	C (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc.	Data Sheet
80m3 / 505bbl	4500	5400	3650	68	79.5	8577	DBAF502218	DBAF502276

** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between X5 and X4
*4) From tank bottom to lower part of E-connection

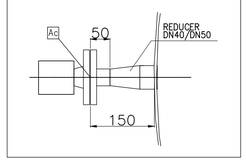
Plates (mm)	80m3	
Roof plate	6	**
Course 1-3	6	**
Bottom plate	8	**

NOTE 1:
- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- ONLY ONE M NOZZLE NEEDED, TO BE PLACED ON DAY TANK SIDE.
- TANK TESTING ACC. TO API 650 (LATEST VERSION)
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION

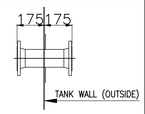
TYPICAL SIDE VIEW OF NOZZLE – A, F & H



CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE – P



REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	--	MNP	--	QRS	

DOCUMENT NUMBER | REFERENCE DOCUMENTS

WÄRTSILÄ

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Energy Business

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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REV.	DATE	MADE	APPVD.	MEMO NO.	EXPLANATION
A	08-May-2020	gbh002	gbh002		Issued for Approval
B	18-May-2020	gbh002	gbh002		Issued for Approval

DRAWING NUMBER	DBAF501379	REVISION b
DATE	18-Mar-2020	DATE 15-Apr-2020 DATE 22-Apr-2020
DRAWN	UKAX01	CHECKED APPROVED
LARSEN & TOUBRO DRAWING NUMBER	019045_XXX_X_X_XXXXX_XXX	REVISION X
OUTOTEC DRAWING NUMBER	900707 _ XXXXX_ XXXX_ XXXXX_ XXXXX	DOCUMENT ID XXXXXXXXXXXX REVISION X
DRAWN	--	--
CHECKED	MNP	XX/XX/XXXX
ENGINEER	--	--
APPROVED	QRS	XX/XX/XXXX
EMPLOYER		
SCALE	1:XX	A1

MANSOURAH - MASSARAH GOLD PROJECT
TITLE-2
TITLE-3
TITLE-4

DRAWING NUMBER MD-216-XXXX-XX-XX-XXX-XXXX REV A0X

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	30-07-2020	24-08-2020	26-10-2023				Date	30-07-2020	24-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 3 GA DRAWING OF LFO DAY TANK

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: LFO Day Tank
49PR POWER GENERATION
Dimensional drawing

Document Number:
Revision:
Revision Date:
Status:

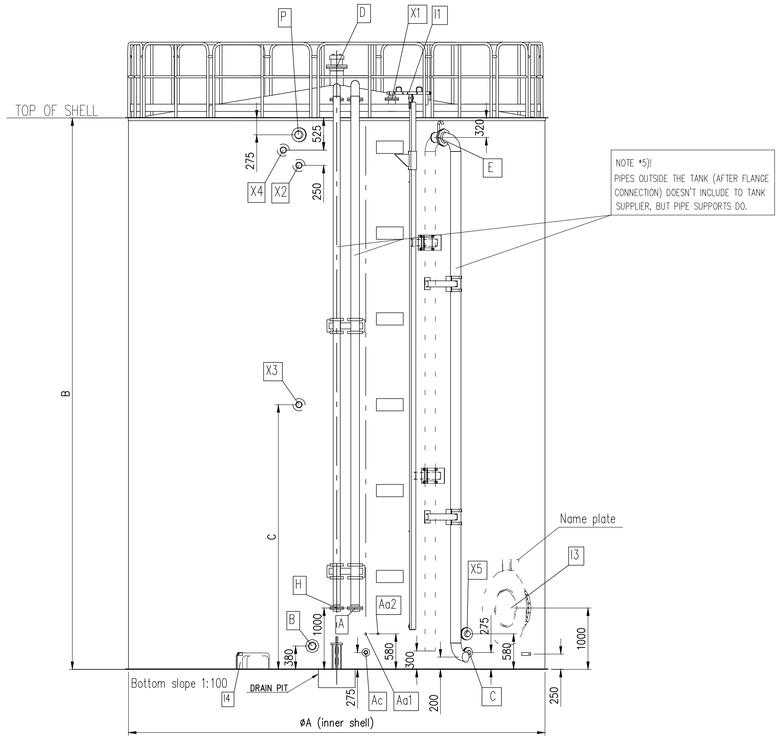
Outotec Document Number:							
L&T Document Number:							

WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF499260 a
	Original Doc no:
	Wartsila Supplier:

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX
REV: A0X



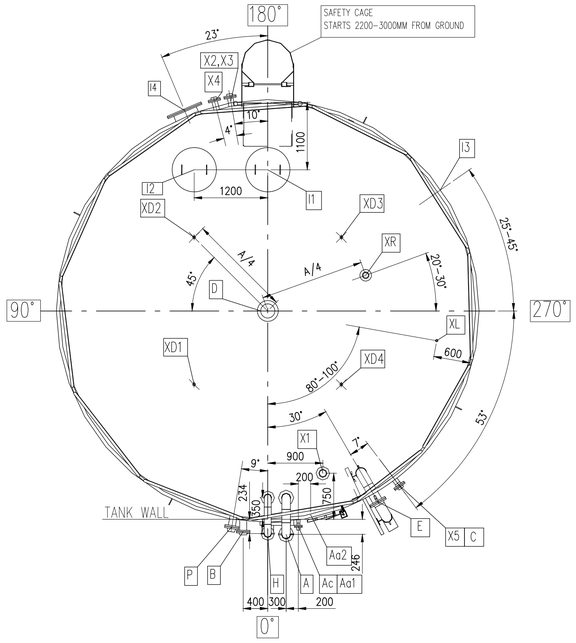
NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	LFO FILLING	5" / DN125	X	300 FROM 0°	150	ROOF	SEE NOTE *5)
B	SUCTION TO UNIT	4" / DN100	X	400 FROM 0°	175	380	
C	DRAIN / EMPTYING	2" / DN50	X	323°	150	275	
D	VENTING PIPE / FLAME ARRESTER	8" - 150lbs	X	AT CENTER	150	ROOF	
E	OVERFLOW	6" / DN150	X	330°	200	320 FROM T.O.S.	SEE DWG DBAD550703 & NOTE *5)
H	RETURN FROM UNIT	4" / DN100	X	0°	150	ROOF	SEE NOTE *5)
I1	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	API 650	180°	250	ROOF	
I2	ROOF MANHOLE/ EMERGENCY VALVE	24" / DN600	API 650	152°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	API 650	225° - 245°	150	1000	
I4	CLEANOUT DOOR	203 X 406		157°		0	See Table 5.9a, 5.10a, 5.11a in API650
Aa1	TEMPERATURE LOCAL	G1/2"		500 FROM 0°		580	
Aa2	TEMPERATURE SENSOR (OPTIONAL / PLUGGED)	G3/4"		700 FROM 0°		580	
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	4" / DN100	X	9°	175	275 FROM T.O.S.	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR		X	900 FROM 0°	150	ROOF	
X2	HIGH LEVEL CONTROL	3" (150 lbs)	X	170°	175 MAX	775 FROM T.O.S.	
X3	LOW LEVEL CONTROL	3" (150 lbs)	X	170°	175 MAX	C	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	X	166°	175 MAX	525 FROM T.O.S.	
X5	LOW LOW LEVEL CONTROL	3" (150 lbs)	X	323°	175 MAX	580	
Ac	CONNECTION FOR PRESSURE TRANSMITTER	1 1/2" (150 lbs)	X	500 FROM 0°	150	275	
XR	ALT. CONN. FOR LEVEL MEASUREMENT(BLIND FLANGE)	3" (150 lbs)	X	240°-250°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL (PLUGGED)	G1"		260°-280°		ROOF	
XD1-XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14NPT		45°, 135°, 225°, 315°		ROOF	TOTAL 4 PCS
x ADDITIONAL INFO FROM API 650 DATA SHEETS							

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
LADDER/STAIRS		1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000		2 SET	
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x ADDITIONAL INFO FROM API 650 DATA SHEETS			

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS



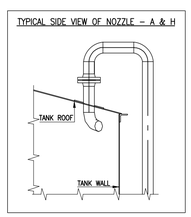
MEASURES				VOLUMES		PRELIMINARY DESIGN VALUES		
Nominal size	ØA (mm)	B (mm)	C (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc.	Data Sheet
150m3 / 1163bbl	6800	5400	2600	155.9	181.5	12765	DBAF502228	DBAF502279

** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between X5 and X4
*4) From tank bottom to lower part of E-connection

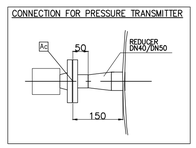
Plates (mm)	150m3
Roof plate	8 **
Course 1-3	6 **
Bottom plate	8 **

NOTE:
- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION
- TANK TESTING ACCORDING TO API 650 (LATEST VERSION)

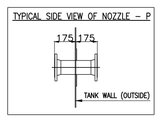
TYPICAL SIDE VIEW OF NOZZLE - A & H



CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE - P



REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	--	MNP	--	QRS	

DOCUMENT NUMBER | REFERENCE DOCUMENTS

WÄRTSILÄ

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DRAWING NUMBER: DBAF499260 | REVISION: XX

DATE	09-Mar-2020	DATE	03-Apr-2020	DATE	22-Apr-2020
DRAWN	UKAX01	CHECKED		APPROVED	

LARSEN & TOUBRO DRAWING NUMBER: 019045_XXX_X_X_XXXXX_XXX | REVISION: X
OUTOTEC DRAWING NUMBER: 900707 __XXXXX_ XXXX_XXXXX_ XXXXX | DOCUMENT ID: XXXXXXXXXXX | REVISION: X

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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DRAWN	--	--
CHECKED	MNP	XX/XX/XXXX
ENGINEER	--	--
APPROVED	QRS	XX/XX/XXXX
EMPLOYER		
SCALE	1:XX	A1

MANSOURAH - MASSARAH GOLD PROJECT
TITLE-2
TITLE-3
TITLE-4

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX | REV: A0X

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	30-07-2020	24-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	30-07-2020	24-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

ATTACHMENT 4 GA DRAWING OF HFO DAY TANK

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: HFO Day Tank
49PR POWER GENERATION
Dimensional drawing

Document Number:
Revision:
Revision Date:
Status:

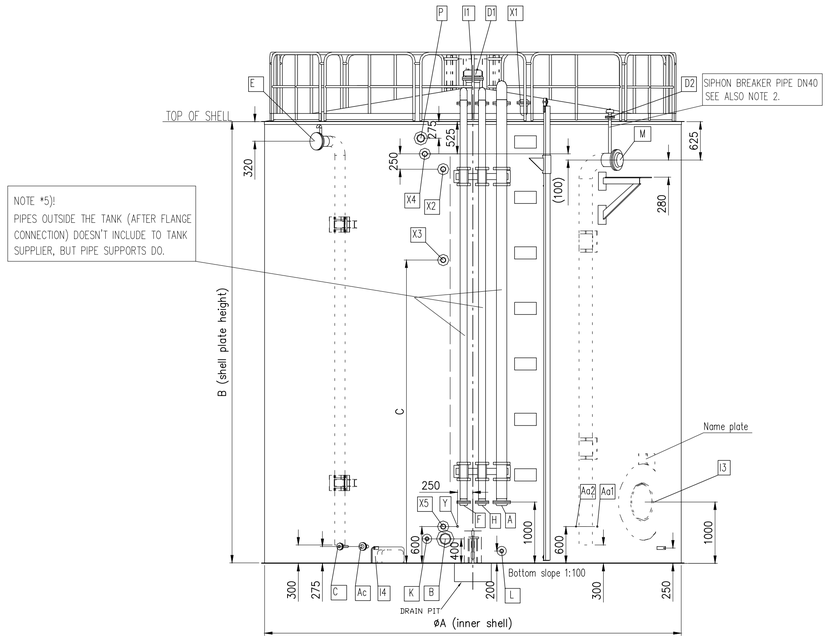
Outotec Document Number:							
L&T Document Number:							

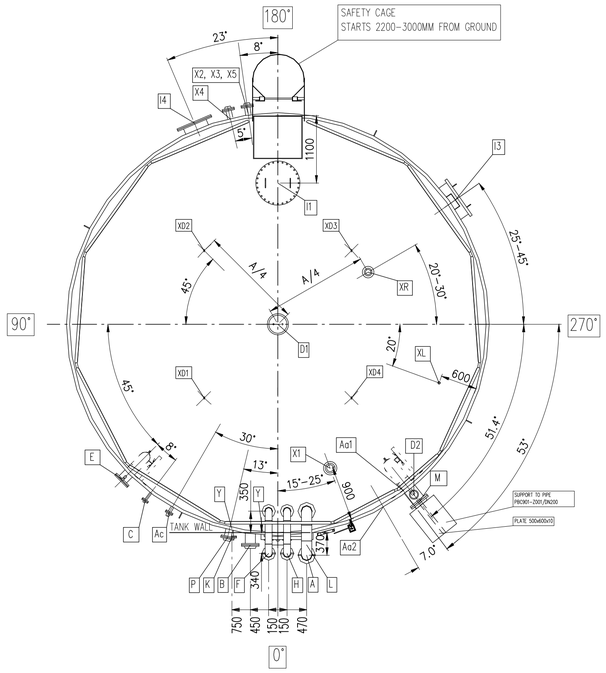
WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF501381 b
	Original Doc no:
	Wartsila Supplier:

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX
REV: A0X





NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	HFO FILLING	5" / DN125	x	470 FROM 0°	150	ROOF	SEE NOTE *5)
B	SUCTION TO UNIT	6" / DN150	x	450 FROM 0°	200	400	
C	DRAIN / EMPTYING	1" / DN25	x	37°	150	275	
D1	VENTING PIPE / FLAME ARRESTER	8", 150lbs	x	AT CENTER	150	ROOF	
D2	SIPHON BREAKER PIPE / FLAME ARRESTER	1 1/2", 150lbs	x	321.4°	400		
E	OVERFLOW	6" / DN150	x	45°	200	320 FROM T.O.S.	SEE NOTE 2, NOTE *5) & DBAB685158
F	RETURN FROM UNIT	4" / DN100	x	150 FROM 0°	150	ROOF	SEE NOTE *5)
H	RETURN FROM UNIT	4" / DN100	x	150 FROM 0°	150	ROOF	SEE NOTE *5)
I1	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	x API 650	180°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	x API 650	225° – 245°	150	1000	
I4	CLEANOUT DOOR	203 X 406		157°		0	See Table 5.9a, 5.10a, 5.11a in API650
K	STEAM INLET	1½" / DN40	x	750 FROM 0°	150	400	MIN. COIL LENGTH 26.3M
L	CONDENSATE OUTLET	1½" / DN40	x	470 FROM 0°	150	200	
M	OVERFLOW TO BUFFER TANK	8" / DN200	x	321.4°	300	625 FROM T.O.S.	SEE WDAAA203547, OVERFLOW SUCTION PIPE, NOTE: THE PIPE SUPPORT TO PBC901-Z001/DN200
Y	STEAM CONTROL THERMOSTAT	G1"		250 FROM 0°		600	
Aa1	TEMPERATURE LOCAL	G 1/2"		323°		600	
Aa2	TEMPERATURE TRANSMITTER	G 1/2"		330°		600	
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	4" / DN100	x	13°	175	275 FROM T.O.S.	
X2	HIGH LEVEL CONTROL (BLIND FLANGE)	3" (150 lbs)	x	172°	175 MAX	775 FROM T.O.S.	
X3	LOW LEVEL CONTROL (BLIND FLANGE)	3" (150 lbs)	x	172°	175 MAX	C	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	x	167°	175 MAX	525 FROM T.O.S.	
X5	LOW LOW LEVEL CONTROL	3" (150 lbs)	x	172°	175 MAX	600	
Ac	CONN.FOR PRESSURE TRANSM.	1½" (150 lbs)	x	30°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR	–	x	335°–345°	150	ROOF	
XR	ALT. CONN. FOR LEVEL MEASUREMENT (BLIND FLANGE)	3" (150 lbs)	x	240°–250°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL (PLUGGED)	G1"		270°–290°		ROOF	
XD1–XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14NPT		45°, 135°, 225°, 315°		ROOF	TOTAL 4 PCS
x	ADDITIONAL INFO FROM API 650 DATA SHEETS						

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
LADDERS	–	1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000		1 SET	
INSULATION	S=100mm	1 SET	TANK SHELL TO BE INSULATED
HEATING COIL	DN40	1 SET	MIN. COIL LENGTH 26.3M
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x ADDITIONAL INFO FROM API 650 DATA SHEETS			

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS

MEASURES				VOLUMES		PRELIMINARY DESIGN VALUES		
Nominal size	øA (mm)	B (mm)	C (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc.	Data Sheet
150m3 / 945bbl	6800	5400	3550	155.2	181.5	14 078	DBAF502219	DBAF502277

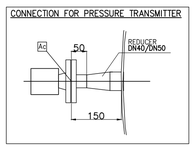
** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between X5 and X4
*4) From tank bottom to lower part of E-connection

Plates (mm)	150m3
Roof plate	8 **
Course 1–3	6 **
Bottom plate	8 **

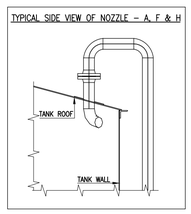
NOTE 1:
- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- TANK TESTING ACC. TO API 650 (LATEST VERSION)
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION

NOTE 2:
- WHEN BOTH TANKS (DAY+BUFFER) ARE USED IN TANK AREA, THEN DAY TANK E CONNECTION IS BLINDED AND IN BUFFER TANK THERE IS OVERFLOW PIPE.

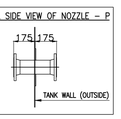
CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE – A, F & H



TYPICAL SIDE VIEW OF NOZZLE – P



REV	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
QRS		A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	--	MNP	--	QRS	

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REV.	DATE	MADE	APPVD.	MEMO NO.	EXPLANATION
A0X	08-May-2020	gbh002	gbh002		

DRAWING NUMBER	DBAF501381	REVISION	b
DATE	18-Mar-2020	DATE	15-Apr-2020
DRAWN	UKAX01	CHECKED	
DATE	22-Apr-2020	APPROVED	

LARSEN & TOUBRO DRAWING NUMBER	019045_XXX_X_X_XXXXX_XXX	REVISION	X
OUTOTEC DRAWING NUMBER	900707 _XXXXX_ XXXX_XXXXX_ XXXXX	DOCUMENT ID	XXXXXXXXXXX
		REVISION	X

DRAWN	--	--	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	MNP	XX/XX/XXXX	TITLE-2
ENGINEER	--	--	TITLE-3
APPROVED	QRS	XX/XX/XXXX	TITLE-4
EMPLOYER			
SCALE	1:XX	A1	

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX
REV: A0X

File name: 2020_ukax01_gbh002_gbh002

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	30-07-2020	24-08-2020	26-10-2023				Date	30-07-2020	24-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 5 GA DRAWING OF SLUDGE TANK & LO STORAGE TANK

Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title:	**Lube Oil Storage Tank** **49PR POWER GENERATION** **Dimensional drawing**

Document Number:
Revision:
Revision Date:
Status:

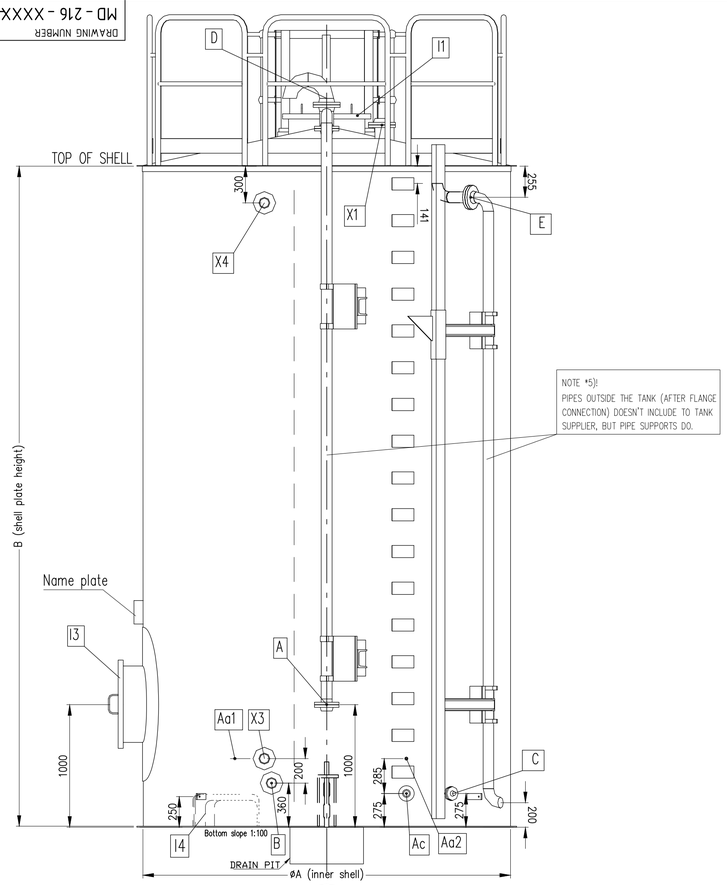
Outotec Document Number:							
L&T Document Number:							

WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF501433 a
	Original Doc no:
	Wartsila Supplier:

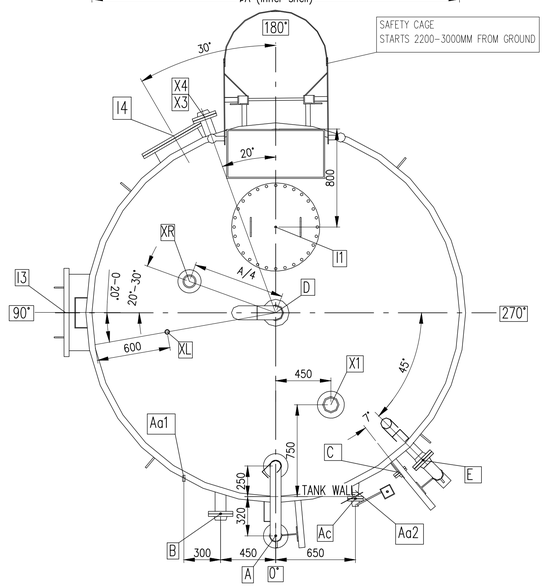
REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX
REV: A0X



NOTE *5):
PIPES OUTSIDE THE TANK (AFTER FLANGE CONNECTION) DOESN'T INCLUDE TO TANK SUPPLIER, BUT PIPE SUPPORTS DO.



NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	LO FILLING	3" / DN80	x	0°	150	ROOF	SEE NOTE *5)
B	SUCTION TO UNIT	3" / DN80	x	450 FROM 0°	175	360	
C	DRAIN / EMPTYING	1" / DN25	x	322°	150	275	
D	VENTING PIPE	4" / DN100	x	AT CENTER	150	ROOF	WIRE MESH
E	OVERFLOW	3" / DN80	x	315°	175	255 FROM T.O.S.	SEE NOTE *5)
I1	ROOF MANHOLE	24" / DN600	x API 650	180°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	x API 650	90°	150	1000	
I4	CLEANOUT DOOR	203 X 406		150°		0	See Table 5.9a, 5.10a, 5.11a in API650
Aa1	TEMPERATURE LOCAL	G1/2"		750 FROM 0°		560	
Aa2	TEMPERATURE SENSOR (OPTIONAL / PLUGGED)	G3/4"		650 FROM 0°		560	
X3	LOW LEVEL CONTROL	3" (150 lbs)	x	160°	175 MAX	560	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	x	160°	175 MAX	300 FROM T.O.S.	
Ac	CONN. FOR PRESSURE TRANSMITTER	1 1/2" (150 lbs)	x	650 FROM 0°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR		x	450 FROM 0°	150	ROOF	
XR	ALT. CONN. FOR LEVEL MEASUREMENT (BLIND FLANGE)	3" (150 lbs)	x	110°-120°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL (PLUGGED)	G1"		70°-90°		ROOF	
x	ADDITIONAL INFO FROM API 650 DATA SHEETS						

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
LADDERS		1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x	ADDITIONAL INFO FROM API 650 DATA SHEETS		

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS

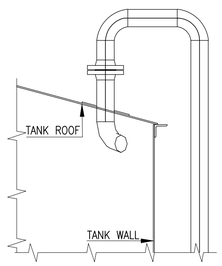
MEASURES			VOLUMES		PRELIMINARY DESIGN VALUES		
Nominal size	øA (mm)	B (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc.	Data Sheet
35m3 / 220bbl	3000	5400	32.1	37.2	4600	DBAF502229	DBAF502281

Plates (mm)	35m3
Roof plate	6 **
Course 1-3	6 **
Bottom plate	8 **

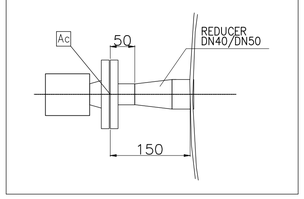
** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between X3 and X4
*4) From tank bottom to overflow pipe (top of elbow)

NOTE:
- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION
- TANK TESTING ACCORDING TO API 650 (LATEST VERSION)

TYPICAL SIDE VIEW OF NOZZLE – A



CONNECTION FOR PRESSURE TRANSMITTER



REV	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
QRS			A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	--	MNP	--	QRS	

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Saudi Arabian Mining Company شركة معادن العربية السعودية
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DRAWING NUMBER	DBAF501433	REVISION	a
DATE	18-Mar-2020	DATE	03-Apr-2020
DRAWN	UKAX01	CHECKED	
DATE		APPROVED	22-Apr-2020

LARSEN & TOUBRO DRAWING NUMBER	019045_XXX_X_X_XXXXX_XXX	REVISION X
OUTOTEC DRAWING NUMBER	900707 __XXXXX_ XXXX_XXXXX_ XXXXX	DOCUMENT ID XXXXXXXXXXX / REVISION X

DRAWN	--	--
CHECKED	MNP	XX/XX/XXXX
ENGINEER	--	--
APPROVED	QRS	XX/XX/XXXX
EMPLOYER		
SCALE	1:XX	A1

MANSOURAH – MASSARAH GOLD PROJECT
TITLE-2
TITLE-3
TITLE-4

DRAWING NUMBER	REV
MD-216-XXXX-XX-XX-XXX-XXXX	A0X

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: Sludge Tank
49PR POWER GENERATION
Dimensional drawing

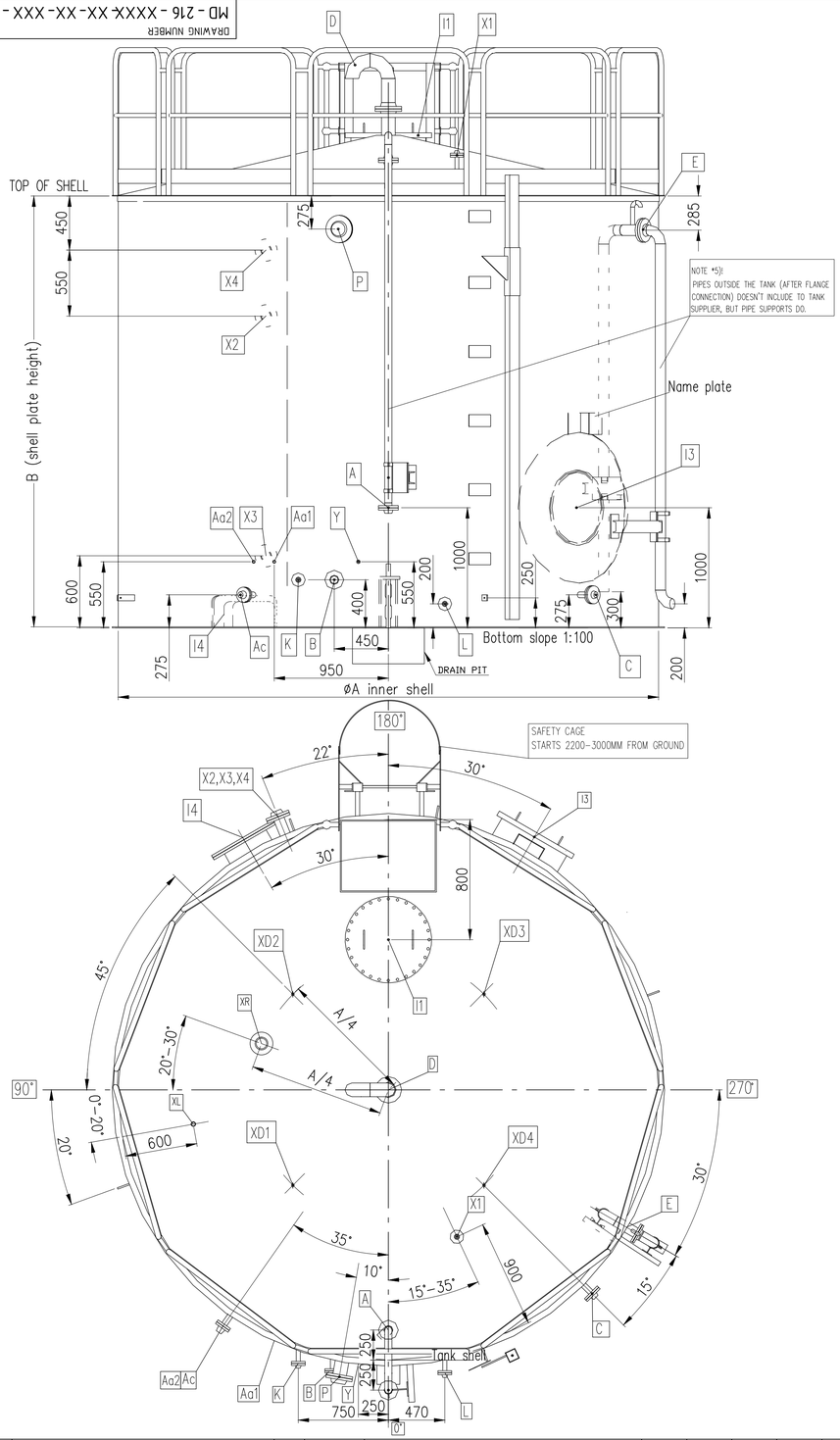
Document Number:
Revision:
Revision Date:
Status:

Outotec Document Number:							
L&T Document Number:							

WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF501382 a
	Original Doc no:
	Wartsila Supplier:

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer



NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	SLUDGE IN	2" / DN50	x	0°	150	ROOF	SEE NOTE *5)
B	SLUDGE OUT	2½" / DN65	x	450 FROM 0°	175	400	
C	DRAIN / EMPTYING	1½" / DN40	x	315°	150	275	
D	VENTING PIPE	4" / DN100	x	AT CENTER	150	ROOF	WIRE MESH
E	OVERFLOW	3" / DN80	x	300°	175	285 FROM T.O.S.	SEE NOTE *5)
I1	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	x API 650	180°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	x API 650	210°	150	1000	
I4	CLEANOUT DOOR	203 X 406		150°		0	See Table 5.9a, 5.10a, 5.11a in API650
K	STEAM INLET	1" / DN25	x	750 FROM 0°	150	400	MIN COIL LENGTH 10.5M
L	CONDENSATE OUTLET	1" / DN25	x	470 FROM 0°	150	200	
Y	STEAM CONTROL THERMOSTAT	G1"		250 FROM 0°		550	
Aa1	TEMPERATURE LOCAL	G1/2"		950 FROM 0°		550	
Aa2	TEMPERATURE SENSOR	G1/2"		35°		550	
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	4" / DN100	x	10°	175	275 FROM T.O.S.	
X2	HIGH LEVEL CONTROL	3" (150 lbs)	x	158°	175 MAX	1000 FROM T.O.S.	
X3	LOW LEVEL CONTROL	3" (150 lbs)	x	158°	175 MAX	600	
X4	HIGH HIGH LEVEL CONTROL	3" (150 lbs)	x	158°	175 MAX	450 FROM T.O.S.	
Ac	CONN.FOR PRESSURE TRANSM. (OPT./BLIND FL.)	1 1/2" (150 lbs)	x	35°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR		x	325°-345°	150	ROOF	
XR	ALT. CONN. FOR LEVEL MEASUREMENT (BLIND FLANGE)	3" (150 lbs)	x	110°-120°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL(PLUGGED)	G1"		70°-90°		ROOF	
XD1-XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14NPT		45°, 135°, 225° AND 315°		ROOF	TOTAL 4 PCS
x ADDITIONAL INFO FROM API 650 DATA SHEETS							

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1070 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
LADDERS		1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000		1 SET	
INSULATION	S=100mm	1 SET	TANK SHELL TO BE INSULATED
HEATING COIL	DN25	1 SET	MIN COIL LENGTH 10.5M
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x ADDITIONAL INFO FROM API 650 DATA SHEETS			

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS

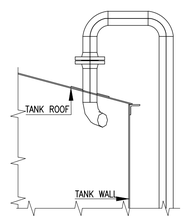
MEASURES			VOLUMES		PRELIMINARY DESIGN VALUES		
Nominal size	ø A (mm)	B (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc.	Data Sheet
35m3 / 220bbl	3000	5400	30.9	36	4700	DBAF502230	DBAF502282

** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between X3 and X4
*4) From tank bottom to lower part of E-connection

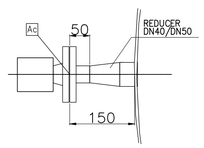
Plates (mm)	35m3
Roof plate	6 **
Course 1-3	6 **
Bottom plate	8 **

NOTE:
- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION
- TANK TESTING ACC. TO API 650 (LATEST VERSION)

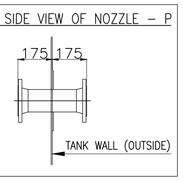
TYPICAL SIDE VIEW OF NOZZLE – A



CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE – P



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
QRS		A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	--	MNP	--	QRS	

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DRAWING NUMBER	DBAF501382	REVISION a
DRAWN	UKAX01	DATE 18-Mar-2020
CHECKED		DATE 03-Apr-2020
APPROVED		DATE 22-Apr-2020

LARSEN & TOUBRO DRAWING NUMBER	019045_XXX_X_X_XXXXX_XXX	REVISION X
OUTOTEC DRAWING NUMBER	900707__XXXXX_XXXX_XXXXX_XXXXX	DOCUMENT ID XXXXXXXXXXX / REVISION X

DRAWN	--	--
CHECKED	MNP	XX/XX/XXXX
ENGINEER	--	--
APPROVED	QRS	XX/XX/XXXX
EMPLOYER		
SCALE	1:XX	A1

MANSOURAH – MASSARAH GOLD PROJECT
TITLE-2
TITLE-3
TITLE-4

DRAWING NUMBER MD-216-XXXX-XX-XX-XXX-XXXX
REV A0X

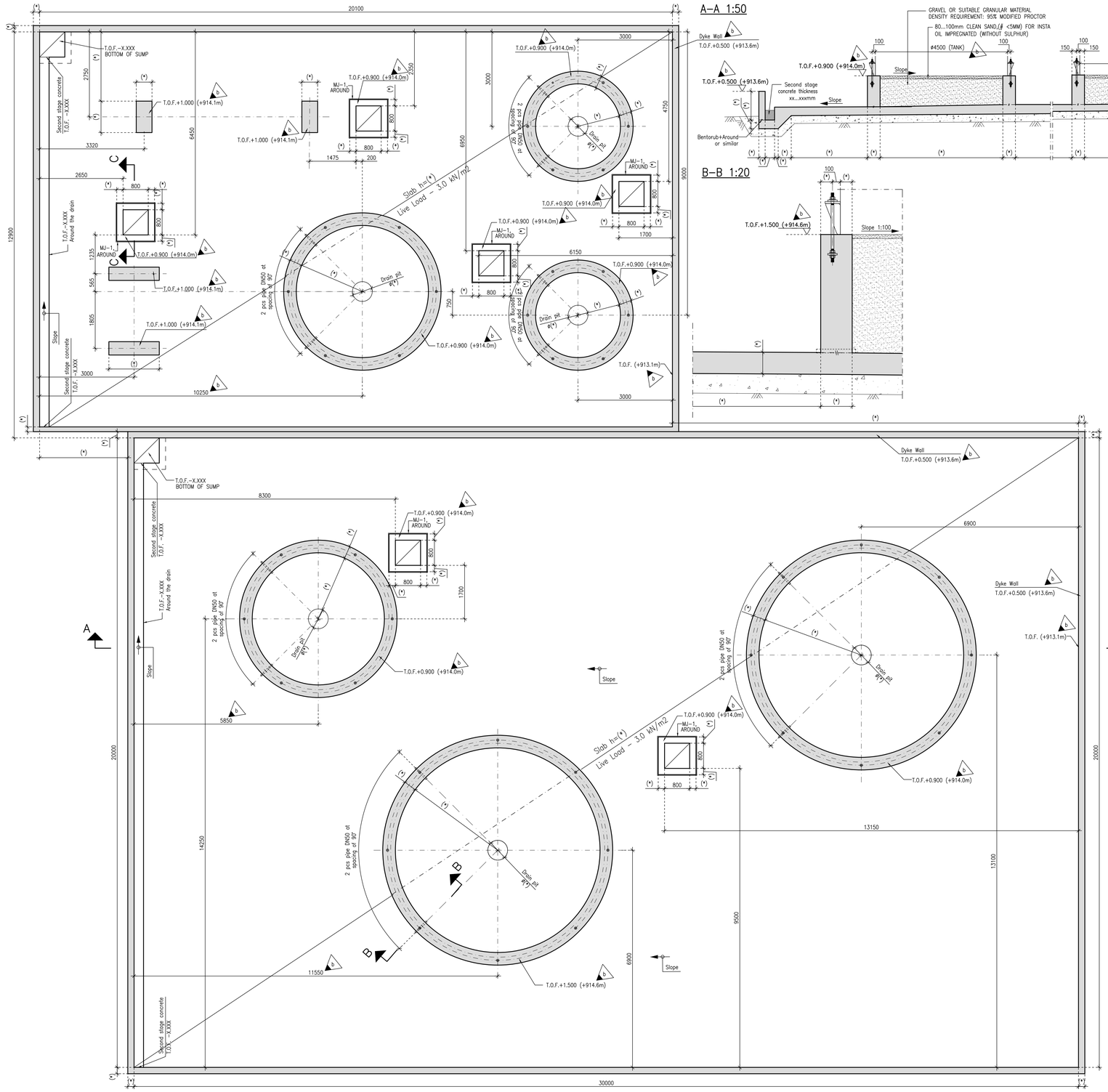
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	30-07-2020	24-08-2020	26-10-2023				Date	30-07-2020	24-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

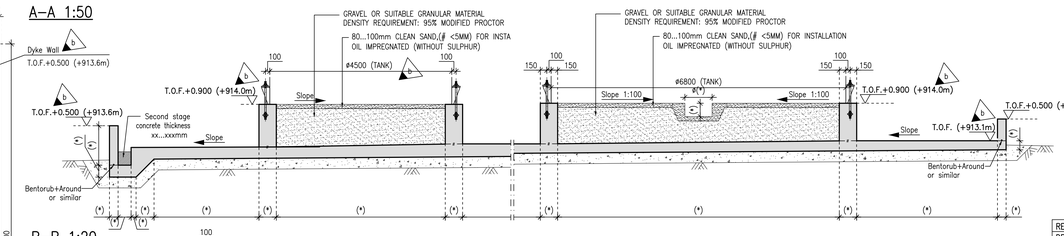
ATTACHMENT 6 DAY TANK AREA LAYOUT DRAWING

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX
REV: A0X

PLAN 1:50



A–A 1:50



B–B 1:20

NOTE:
1) This drawing shall be mainly used for dimensional information purposes. Final construction drawing which concerns reinforcing and possible piling of the foundation to be done by the civil designer in accordance with the soil investigations at site. However below are given some points made by Wartsila.
2) Beam size and location according to final design
3) MJ location according to final design
4) Conduit locations according to Cable Conduit design.
5) Drain points according to Unit Location drawing
6) Location and size of the drain channels to be checked according to final design

NOTE! LOCATION AND TOP OF FOUNDATION LEVEL T.O.F. ACCORDING TO FOUNDATION LOCATION DRG.

REFERENCE DRAWINGS	
DRG NO.	NAME OF THE DRAWING
DBAF518690	FOUNDATION LOCATION PLAN – BY CUSTOMER
DBAF491079	UNIT LOCATION DRAWING DAY TANK AREA
DBAF508451	OILY WATER DRAINAGE DAY TANK AREA
DBAF490344	DOCUMENT AND FLOW DIAGRAM NUMBERS TANK LIST
DBAF510203	SITE LOAD PLAN DRAWING

MATERIAL	AMOUNT
Concrete	**.* m3
Formwork	**.* m2
Bentorub+	**.* m
Second stage concrete	**.* m3
Polymeric sealant, oil proof type	**.* m
Sealing strip	**.* m
Separating material t=12 mm, h=250mm	**.* m
Separating material t=12 mm, h=200mm	**.* m
Inspection pipe: ø50mm	**.* m
Anchor bolt: M36, L=880	**.* Pcs
HILTI HST-R M20/200x60 Stud Anchor	**.* Pcs
Waterstop JOCO 200	**.* m
L-profile 50x50x5	*.* m
Grooved steel sheet, t= 4 mm, 1100x1100	* pcs

TOLERANCE FOR LEVEL: TOP OF RINGWALL
(ACCORDING TO API 650)
TOP OF THE RINGWALL SHALL BE LEVEL WITHIN ± 1/8 INCH (3 mm) IN ANY 30 FEET (9000 mm) OF THE CIRCUMFERENCE AND ± 1/4 INCH (6 mm) IN THE TOTAL CIRCUMFERENCE MEASURED FROM THE AVERAGE ELEVATION.

ANCHOR BOLTS PROJECTION TOLERANCES SHALL BE +1/4" / -0" AND SHALL BE WITHIN 1/8" OF THEORETICAL LOCATION.

NOTES:
1. TOP OF FOUNDATIONS MARKED ACCORDING TO CUSTOMER'S INPUT
   T.O.F. +0.000 = (FFL) +913.100 m
2. " * "FOUNDATION SIZE AND LOCATION ACCORDING TO FINAL DESIGN.
3. THE BOLT EMBEDMENT LENGTH TO BE CALCULATED BY CIVIL (FOUNDATION) DESIGN.

FOR INFORMATION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

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DRAWING NUMBER	DBAF507554	REVISION b
DATE / DRAWN	9-Jul-2020 / MPAX08	DATE / CHECKED 9-Jul-2020 / NCHX03
DATE / APPROVED	09-Jul-2020 / AKRX06	
LARSEN & TOUBRO DRAWING NUMBER	019045_XXX_X_X_XXXXX_XXX	REVISION X
OUTOTEC DRAWING NUMBER	900707 _XXXXX_ XXXX_XXXXX_ XXXXX	DOCUMENT ID XXXXXXXXXXX REVISION X

DRAWN	---	---
CHECKED	MNP	XX/XX/XXXX
ENGINEER	---	25/11/2019
APPROVED	QRS	XX/XX/XXXX
EMPLOYER		
SCALE	1:50	A0

MANSOURAH - MASSARAH GOLD PROJECT
Dimensional drawing
POWER GENERATION
DAY TANK AREA - INFORMATION DRAWING

DRAWING NUMBER: MD-216-XXXX-XX-XX-XXX-XXXX
REV: A0X

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	30-07-2020	24-08-2020	26-10-2023				Date	30-07-2020	24-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 7 LOAD TABLE FOR STORAGE TANKS DOCUMENT NO. DBAF490344

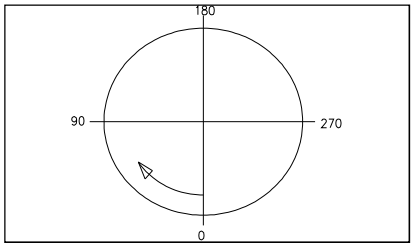
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Power Plants

Doc. Classification: Confidential

Title:	Tank list	**DocID:**	DBAF490344
		Revision:	f
Author:	Urpo Kähtävä	**Sheets:**	1
Approved by:	Shahu Injal / 28.07.2020	**Status:**	Approved
Project:	P/20002 Mansourah - Massarah Gold Mine		
Description:	Document and flow diagram numbers		
Type:	List		



Mansourah - Massarah Gold Mine, Saudi-Arabia, Design acc. to API 650, wind velocity 43m/s, Seismic Zone 1, design pressure 0.65kPa.
Horizontal tanks acc. to EN12285-2.

Tank	Nominal volume [m3]	Document number	Process Flow diagram	Diam.of tank [mm]	Height of shell [mm]	Qty	Net volume (betw. X4 and X5) [m3]	Number of anch.	Place of the first anchor bolt (°)	Diam. Of anch. bolts	Wind Shear Force (N)	Wind moment on tank (Nm)	Seismic Shear Force (N)	Ringwall overturning moment (Nm)	Slab Overturning moment (Nm)
HFO Storage Tank	4000m3	DBAF501378a	DBAF501831a	19200	14400	1	3662.5	24	0	M36	198,500	4,932,040	**1,330,100**	**7,199,480**	**10,490,000**
HFO Buffer Tank	80m3	DBAF501379b	DBAF504726a	4500	5400	1	68	6	0	M36	**41,045**	**116,870**	32,753	72,420	91,484
HFO Day Tank	150m3	DBAF501381c	DBAF504728a	6800	5400	1	155.2	8	0	M36	62,785	183,810	**64,801**	**133,520**	**191,050**
LFO Storage Tank	1000m3	DBAF499249a	DBAF504723a	11500	10800	1	993.5	14	0	M36	83,610	1,196,110	**396,700**	**1,710,690**	**2,259,100**
LFO Day Tank	150m3	DBAF499260b	DBAF504730a	6800	5400	1	155.9	8	0	M36	61,127	179,340	**64,209**	**131,980**	**189,470**
Lube Oil Storage Tank	35m3	DBAF501433b	DBAF502325a	3000	5400	1		6	0	M36	**25,795**	**72,275**	16,521	40,100	46,543
Lube Oil Service Tank	7m3	DBAF501457b	DBAF502380a	1600	length 3737	1		4	0	M20	-	-	-	-	-
Used Lube Oil Storage tank	20m3	DBAF501458b	DBAF504733a	2000	length 6959	1		4	-	M20	-	-	-	-	-
Sludge Tank	35m3	DBAF501382a	DBAF504737a	3000	5400	1		6	0	M36	**27,453**	**76,751**	16,534	38,965	45,217
Oily Water Buffer Tank	50m3	DBAF501459b	DBAF504735a	4500	3600	1		6	0	M36	**27,497**	**53,580**	20,036	27,722	38,559
Treated Water Storage Tank	10m3	DBAF501462b	DBAF504739a	1600	length 5347	1		4	-	M20	-	-	-	-	-

(DBAF454192-), Anchor bolt chair type A for tank size 300m3 and below.
(WDAAA272325b), Anchor bolt chair type B for tank size above 300m3.